

6	(a)	A photon is a discrete <u>amount/quantum/packet</u> of <u>energy</u> of <u>electromagnetic radiation</u> .	B1
		Marker's comment Poorly done. Many did not show the basic understanding of what a photon is. Some failed by only relating photon to the energy needed in photoelectric effect or line spectra. Students need to understand the fundamental concepts, revolving around the energy that the light provides.	
	(b)	The electrons in the gas atoms/molecules (interacts and) <u>absorbs photons</u> and their energy, causing electrons in the gas atoms/molecules to move to a higher energy level or to be <u>excited</u>. (For the excitation to occurs,) the energy of the photons must be the difference in the energy of the electron energy levels of the gas atoms/molecules. When the electrons de-excite (after a random short interval), photons are emitted in <u>all directions</u>. Since the intensity of such re-emitted photons in the original direction are lower than that of the other incoming photons, the spectrum consists of a number of dark lines in coloured spectrum.	B1 B1 B1 B1 A0
		Marker's comment Below satisfactory. Some are mistaken that the diffraction grating is the source of the dark lines and went on to use destructive interference to explain. A few uses the concept of X-ray spectrum to explain but unable to come up with any explanation. Several understand the dark lines are due to the light interacting with the gas molecules. Details may be lacking, e.g. some failed to include the existence of photons to explain while other did not mention how the re-emitted of the photons results in all directions lead to the dark lines in the continuous spectrum.	
	(c)	(i) $E_p = \frac{hc}{\lambda}$ $= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{420 \times 10^{-9}}$ $= 4.736 \times 10^{-19} \text{ J}$ $= 2.96 \text{ eV}$ $E_p = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{740 \times 10^{-9}}$ $= 2.688 \times 10^{-19} \text{ J}$ $= 1.68 \text{ eV}$	M1 A0 M1 A0
		Marker's comment Average. It was surprising that some left this questions unattempted, suggesting that they did not know or understand how to compute energy of photon using wavelength. Additionally, since this is a "show" question, students need to show the full working and subtraction that results in the energy computed in eV.	

		(ii) - 3.4 eV $\rightarrow$ - 1.5 eV - 3.4 eV $\rightarrow$ - 0.85 eV - 3.4 eV $\rightarrow$ - 0.54 eV All correct and none incorrect 2/2 2 correct and 1 incorrect <u>or</u> only 2 correct $\frac{1}{2}$ <u>or</u> 3 correct ΔE but de-excitation	B2
		Marker's comment Below satisfactory. Many did not know that it is possible for gas molecules to be at higher energy levels due to thermal energy. They wrongly thought that the absorption of the photons must occur from the ground state. Some did not understand that the absorption of the photons, hence removal of such photons at these wavelengths, is the cause of the dark lines in the continuous spectrum. Students need to understand that the electrons will only transit from one energy level to another, and not in between.	
	(d)	$p = \frac{h}{\lambda}$ $9.11 \times 10^{-31} \times v = \frac{6.63 \times 10^{-34}}{420 \times 10^{-9}}$ $v = 1730 \approx 1700 \text{ m s}^{-1}$	C1 A1
		Marker's comment Below satisfactory. Many either leave this unattempted or wrongly use the equation $E_p = \frac{hc}{\lambda}$ and then use the energy of the photon to equate to the kinetic energy of the electrons. Some did not use the shortest wavelength of the light beam in the question.	

7	(a)	(i)	LEDs, which is a type of diode, will only produce light when it is forward biased.	P1
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